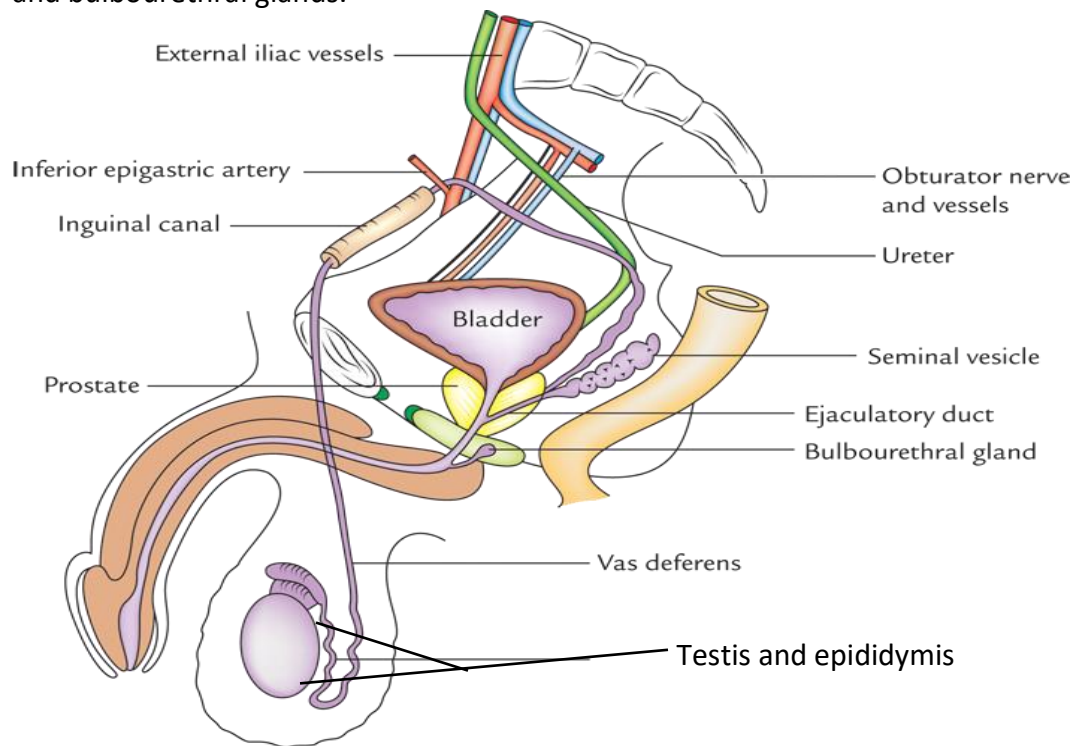


Reproductive system

MALEREPRODUCTIVE SYSTEM

Male reproductive system consists of testis, epididymis, ductus deferens or vasdeferens, ejaculatory ducts and penis. Accessory glandular structures include seminal vesicles, prostate and bulbourethral glands.



TESTIS

It is a male gonad. The function of testis include production of spermatozoa and secretion of testosterone.

Shape : Oval

Location : Testis is suspended in the scrotum by spermatic cord.

External features

The testis consists of

Two poles : Upper and lower

Two borders : Anterior and posterior

Two surfaces : Medial and lateral

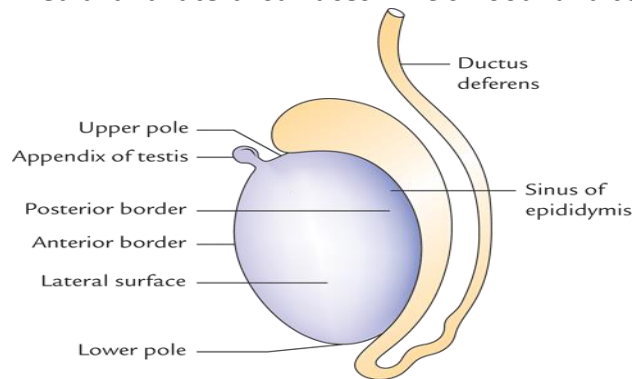
Upper pole : Convex and attached to spermatic cord.

Lower pole : Convex and smooth.

Anterior border : Rounded and covered by tunica vaginalis.

Posterior border : Straight , partly covered by tunica vaginalis and it is attached by epididymis.

Medial and lateral surfaces : Are smooth and convex



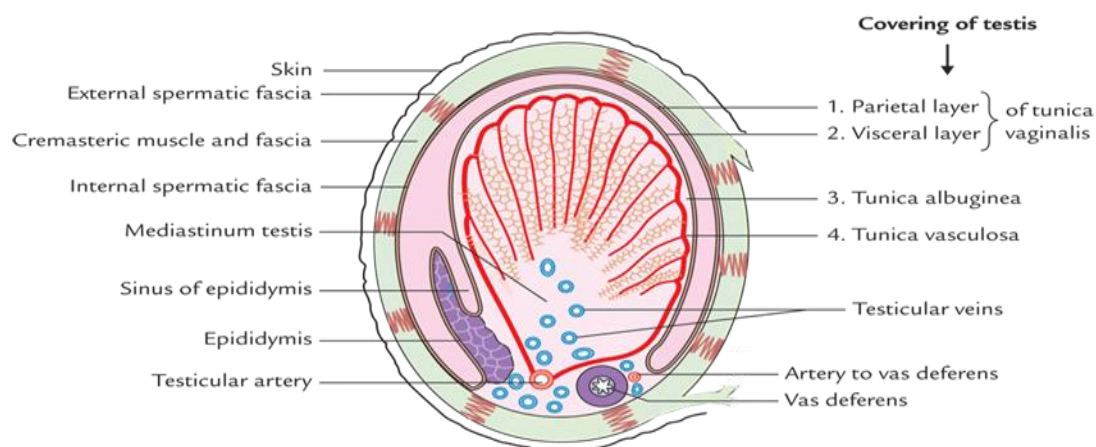
Coverings

It is covered by three coats or layers

1. Tunica vaginalis : It is superficial layer. It is serous sac, and it presents parietal and visceral layers with a potential cavity between them. It covers the whole testis, except for its posterior border.
2. Tunica albuginea : It is the middle layer. It is a thick dense layer of fibrous tissue enclosing the testis. It is covered by the visceral layer of tunica vaginalis, except posteriorly. Posteriorly this layer is thickened to form incomplete vertical septum called mediastinum testis.
3. Tunica vasculosa : It is the innermost vascular layer of the testis, lining the lobules of the testis.

Structure

Testis enclosed in a fibrous capsule, the tunica albuginea. Posteriorly it is thickened to form incomplete vertical septum called mediastinum testis. Numerous septa extend from the mediastinum and divide the interior of testis into 200 – 300 lobules. Each lobule contain 2 to 4 coiled seminiferous tubules, that produces spermatozoa. The thin thread like loops of seminiferous tubules joins each other to form straight tubules. Network of these tubules called rete testis. The small efferent ductules connect the tubules of rete-testis to the upper end of epididymis.



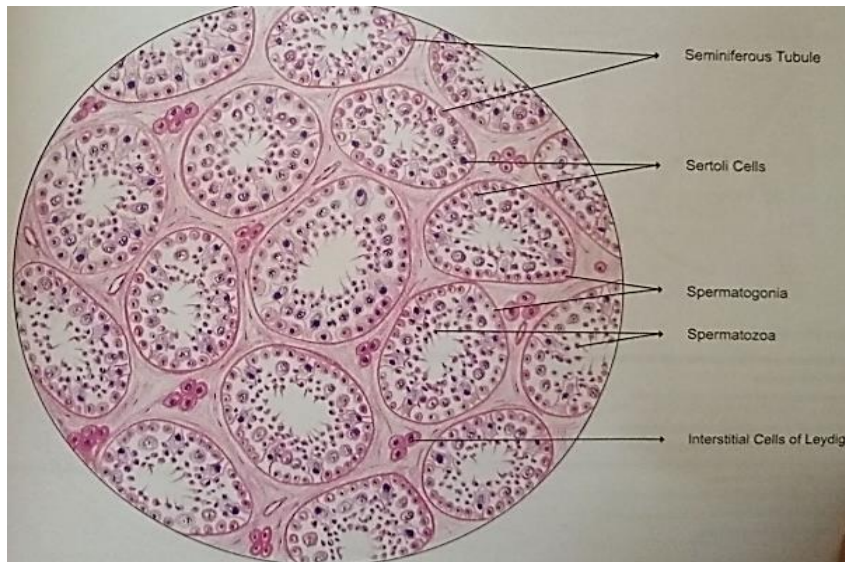
Arterial supply : testicular artery

Venous drainage : pampiniform plexus of veins. Right side drains in to inferior venacava and left side drains in to left renal vein.

Nerve supply : T10 spinal segment

Microscopic structure

Connective tissue covering – tunica albuginea. Seminiferous tubules lined by different stages of spermatogenesis and Sertoli cells. Sertoli cells – tall columnar, Function – support and nutrition, phagocytosis and secretion of hormones. Spermatogonia – germ cells, lie near the basal membrane, undergo mitosis to form primary spermatocytes. Primary spermatocytes – large cells. Each undergoes meiosis to form two secondary spermatocytes. Secondary spermatocytes – smaller, each undergoes meiosis to form two spermatids. They Undergoes morphological changes to form spermatozoa by the process of spermiogenesis. Spaces between the seminiferous tubules contain loose connective tissue with blood vessels. Also contains interstitial cells of Leydig – rounded large cells, secrete testosterone – male sex hormones.



Applied anatomy :

Hydrocele : It is the accumulation of the fluid within the tunica vaginalis.

Varicocele : It is produced by dilatation of the pampiniform plexus of veins.

EPIDIDYMIS

It is a mass made up highly coiled tubes. That act as reservoir of spermatozoa.

Shape : Comma shape

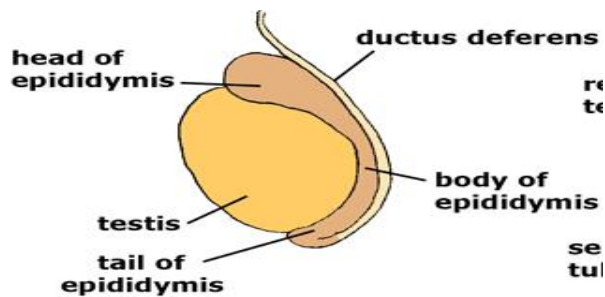
Location : Superior and posterolateral surface of testis

Parts : It is divided into three parts . Head, body and tail

Head : It is enlarged and made up of highly coiled efferent ductules. It is connected to the upper pole of testis by efferent ductules.

Body : It is the middle part. It is made up of single duct.

Tail : The lower part is called tail. It is made up of single duct and becomes continuous with ductus deferens.



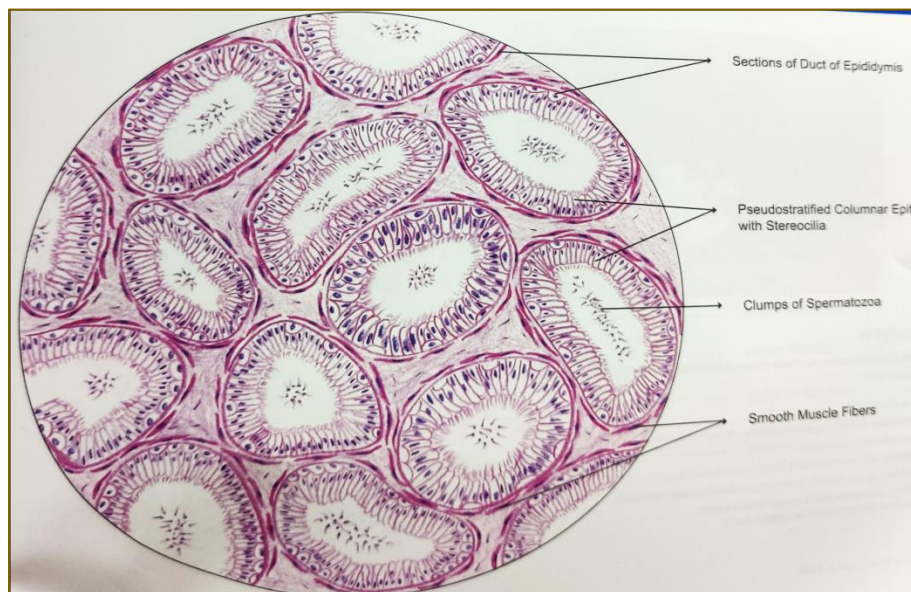
Arterial supply : Testicular artery and artery to the vasdeferens

Venous drainage : Pampiniform Plexus of vein

Nerve supply : T11 to L1 spinal segments.

Microscopic structure

Highly coiled muscular tubule. Coils of tubule held by fibrous connective tissue. Each tubule is lined by pseudostratified columnar epithelium with stereocilia. Each tubule covered by smooth muscles. Sterocilia are large microvilli for absorption of fluids.



DUCTUS DEFERENS OR VASDEFERENS

It is a thick walled muscular tube, which transmits spermatozoa from the epididymis to the ejaculatory duct .

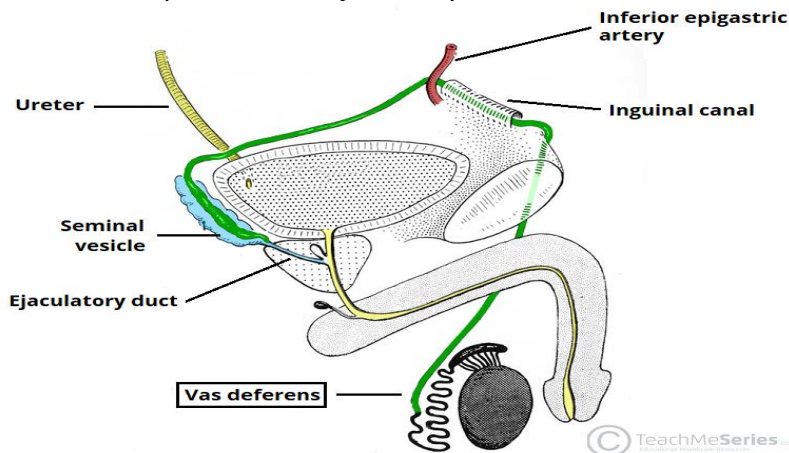
Length : 45cm

Location : Scrotum, inguinal canal, greater and lesser pelvis.

Course and relations

- It is a continuation of the tail of the epididymis.
- Along the posterior border of testis, medial to the epididymis.
- In the spermatic cord – it lies in the posterior part of the spermatic cord. It runs upward to the superficial inguinal ring and then transverse the inguinal canal.
- In the greater pelvis : At the deep inguinal ring it leaves the spermatic cord, and hook round the inferior epigastric artery and enters the lesser pelvis.

- In the lesser pelvis : It crosses the obliterated umbilical artery, obturator nerve and vessels. It then crosses the ureter. Reaching the base of urinary bladder it lies medial to seminal vesicle. At the base of the prostate, it is joined by the duct of the seminal vesicle to form ejaculatory



duct.

Arterial supply: artery to vasdeferens, superior and inferior vesical arteries and middle rectal artery

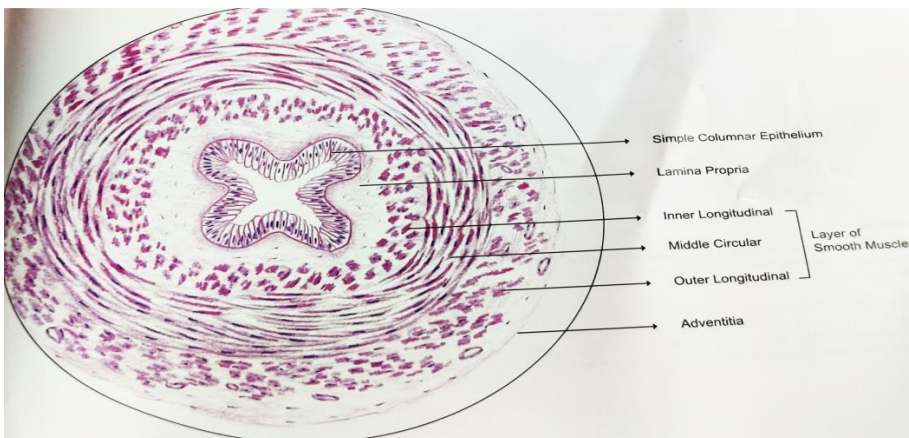
Venous drainage : veins from the ducts join the vesical venous plexus which opens into the internal iliac vein.

Nerve supply : pelvic splanchnic nerve

Microscopic structure

Thick walled muscular tube

1. Mucous membrane shows irregular folds and lined by simple columnar epithelium, become pseudostratified columnar in distal part of duct. Epithelium supported by lamina propria contain loose connective tissue & elastic fibers
2. Thick muscle coat present, consist of smooth muscle. Muscle coat arranged in the form of inner longitudinal, middle circular & outer longitudinal layer.
3. Outer most layer Adventitia, consist of connective tissue.



Applied anatomy

Vasectomy : it is one of the commonest operations being done in india for purposes of family planning.

PROSTATE

It is an accessory gland of male reproductive system. It is a fibromuscular gland.

Shape : pyramidal or inverted cone

Location : in the lesser pelvis, below the neck of urinary bladder.

External features and relations

It consists of apex, base and four surfaces – anterior, posterior and two inferolateral.

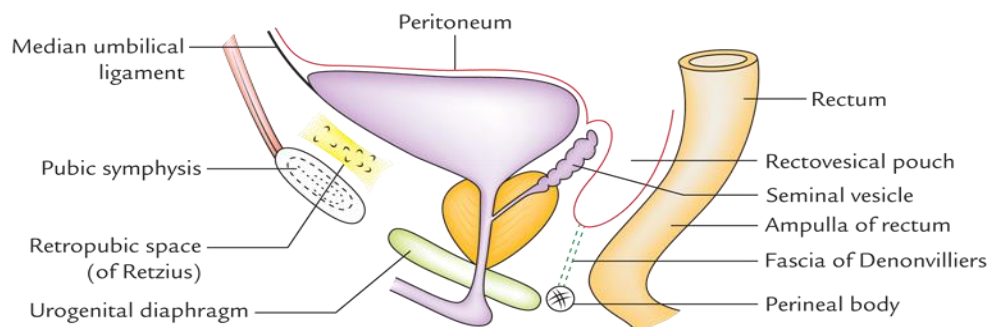
Apex : directed downwards and related to urogenital diaphragm

Base : directed upward and it surrounds the neck of urinary bladder.

Anterior surface : is related by pubic symphysis, lower end of this surface is pierced by the urethra.

Posterior surface : is triangular in shape and this surface is separated from rectum by fascia of Denonvilliers. Its upper border near the median plane on each side pierced by ejaculatory duct.

Inferolateral surface : is related by levator ani muscle.



Lobes of prostate

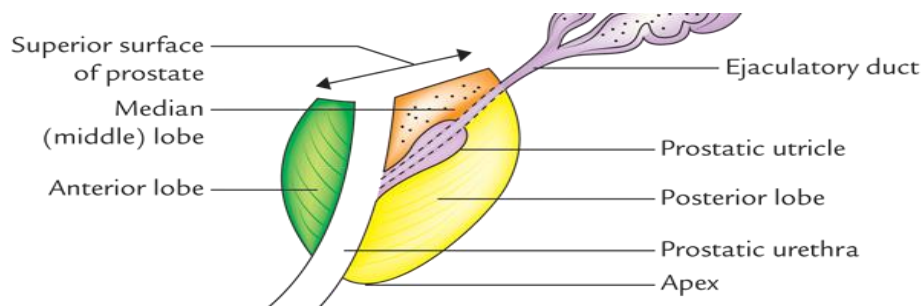
It has 5 lobes. Anterior, posterior, median and right and left lateral

Anterior lobe : is small and lies in front of the urethra. It connects the two lateral lobes. It is devoid of the glandular tissue and often called isthmus.

Posterior lobe : it lies behind the ejaculatory duct and median lobe. It connects the two lateral lobes. Primary carcinoma of the prostate begins in this lobe.

Median lobe : is wedge shaped, lies behind urethra and in front of ejaculatory duct. It produces an elevation in the urinary bladder called uvula vesicae. It contains glandular tissue and is a common site of adenoma.

Lateral lobe : it lies one on each side of urethra. Lateral lobes contain glandular tissue. Adenoma arises in the old age.



Capsule of the prostate

True capsule : It is formed by the condensation of peripheral part of the gland .

False capsule : It lies outside the true capsule and is derived from the pelvic fascia.

Blood supply

Arterial supply : Inferior vesical, middle rectal and internal pudental arteries.

Venous drainage : Vein form a rich plexus and it communicate with vesical plexus and with internal pudental vein and drains into internal iliac vein .

Nerve supply

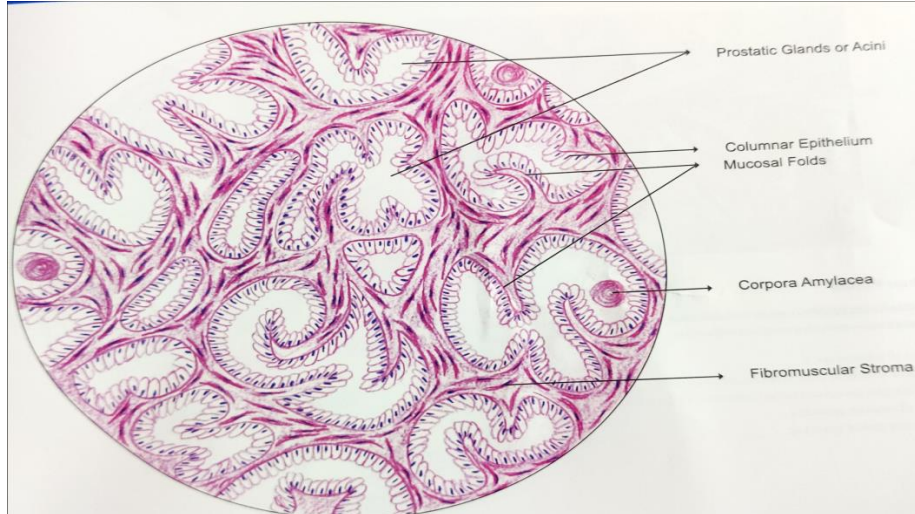
Sympathetic : L1 and L2 spinal segments

Parasympathetic : S2 to S4 spinal segments

Microscopic structure

Prostatic glands (acini/follicles) lined by simple columnar epithelium & epithelium thrown into numerous folds, lumen contain corpora amylacea.(calcified glycoprotein).

Acini seperated by thick fibro- muscular stroma is made up of collagen fibers & smooth muscles



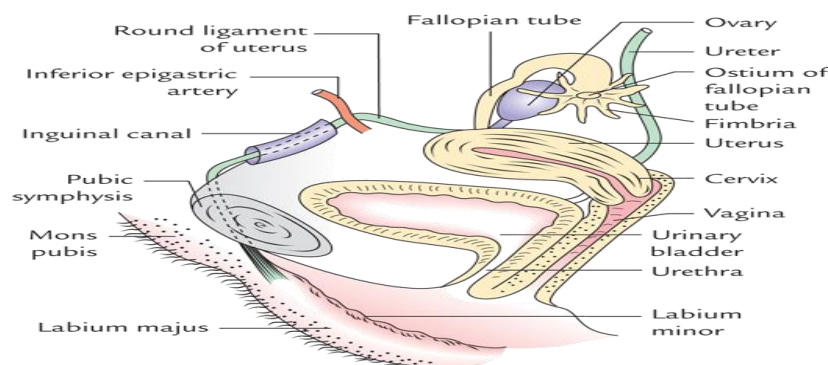
Applied anatomy

Prostatitis : inflammation of prostate called prostatitis

Prostate is a common site of carcinoma

FEMALE REPRODUCTIVE SYSTEM

Female reproductive system consist of ovaries ,uterine tube, uterus, vagina and accessory organs include mons pubis, labia majora, labia minora, clitoris, vestibular bulb, greater vestibular gland, vestibule and mammary gland or breast.



OVARIES

Ovaries are female gonads and produce oocytes.

Shape : almond

Location : ovarian fossa on the lateral pelvic wall.

External features and relations

It consist of :

1. Two poles – upper or tubal and lower or uterine
2. Two borders – anterior and posterior
3. Two surfaces – lateral and medial

Upper pole – it is broader and related to uterine tube and external iliac vein and it is attached to ovarian fimbria and suspensory ligament of the ovary

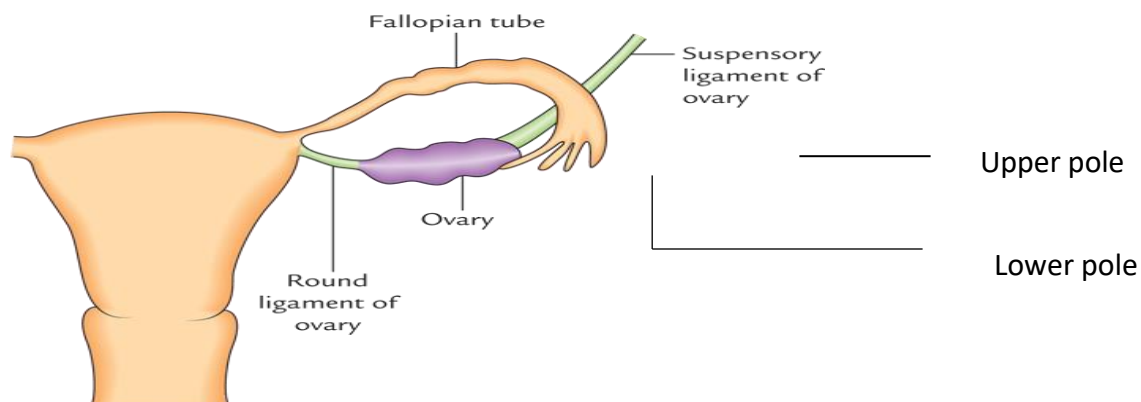
Lower pole - it is narrow and related to uterus. It is connected into uterus by ligament of the ovary.

Anterior border : it is straight and is related to the uterine It is attached to the part of broad ligament called mesovarium. This border is called hilum of ovary.

Posterior border : it is convex and related to the uterine tube and ureter.

Lateral surface : it is related to the ovarian fossa on the lateral pelvic wall.

Medial surface : is covered by uterine tube.



Arterial supply : ovarian and uterine arteries

Venous drainage :pampiniform plexus around the artery drains into inferior venacava and left renal vein.

Nerve supply : sympathetic – T10 and T11 spinal segments

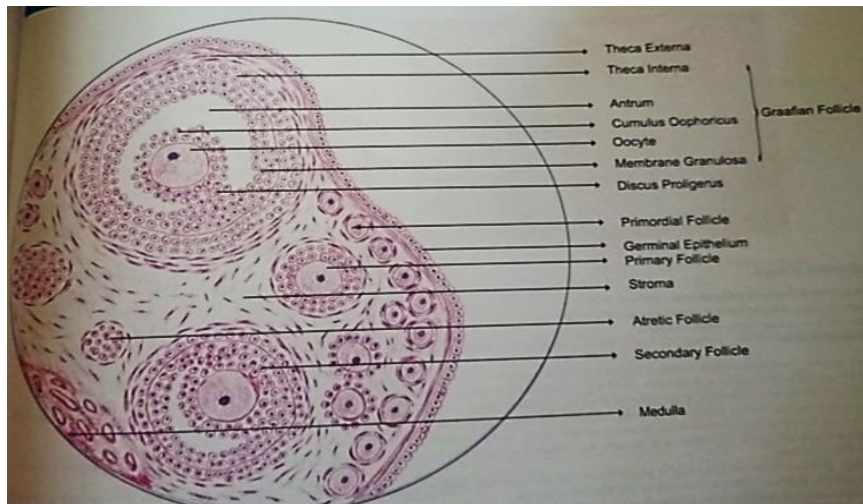
Parasympathetic – S2 to S4 spinal segments

Microscopic structure

Surface lined by germinal epithelium made of cuboidal cells below which lies the connective tissue covering called tunica albuginea. Ovary divides into outer cortex and inner medulla.

Cortex contains follicle in different stages of development separated by the stroma. Cortex contain follicles are Primordial follicle,Primary follicle, Secondary follicle and Graafianfollicle.

Follicular cells secrete estrogen. Medulla – contains connective tissue contain blood vessels and smooth muscles.



Applied anatomy

Ovarian cysts : developmental arrest of ovarian follicles may results in the formation of one or more small ovarian cysts.

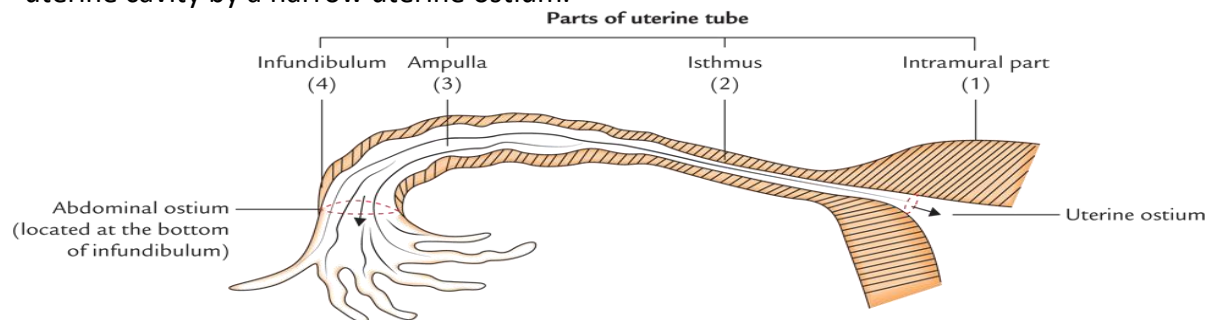
Carcinoma of ovary is common called ovarian cancer.

UTERINE TUBE OR FALLOPIAN TUBE

They are tortuous ducts, which convey ova from ovary to the uterus. It is about 10cm long.

Subdivisions

1. **Infundibulum** : it is the lateral end. It is funnel shaped and about 1cm long. It bears number of finger like processes called fimbria. One of the fimbriae longer and attached to the ovary called ovarian fimbriae. It is opens into the lateral pelvic wall through abdominal ostium.
2. **Ampulla** : it is 5cm long. It lies medial to the infundibulum. It is thin walled, tortuous and widest part. It is site of fertilization of ovum.
3. **Isthmus** : it is about 2.5cm to 3cm long. It is narrow, rounded and cord like. It lies medial to the ampulla.
4. **Intramural part** : it is about 1cm long. It lies with in the wall of the uterus. It opens into the uterine cavity by a narrow uterine ostium.

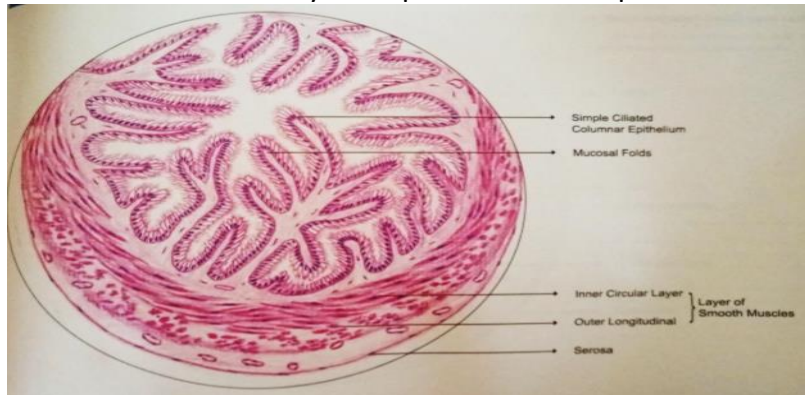


Blood supply : uterine and ovarian arteries. Venous drainage into pampiniform plexus of vein into uterine veins.

Nerve supply : sympathetic – T10 to T12 spinal segments
Parasympathetic – S2 to S4 spinal segments

Microscopic structure

Mucous layer – thrown into numerous branching anastomosing folds which fill the lumen. Lined by columnar cells – cells are of two types: ciliated cells – which help in the movement of ova towards the uterine cavity. Non ciliated cells – secretory or peg cells, secretion help in nourishment of ova. Muscle layer – inner circular and outer longitudinal layer of smooth muscles. Serosa lined by flat squamous cells of peritoneum.



Applied anatomy

Salpingitis : inflammation of uterine tube.

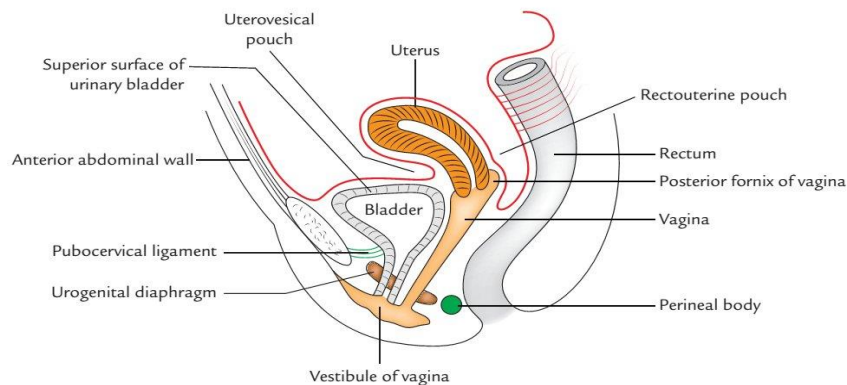
Tubectomy : by removing a segment of the uterine tube on both side for the purposes of family planning.

UTERUS

It is a thick walled hollow muscular organ. It consists of three layers endometrium, myometrium and epimetrium.

Shape : Pear

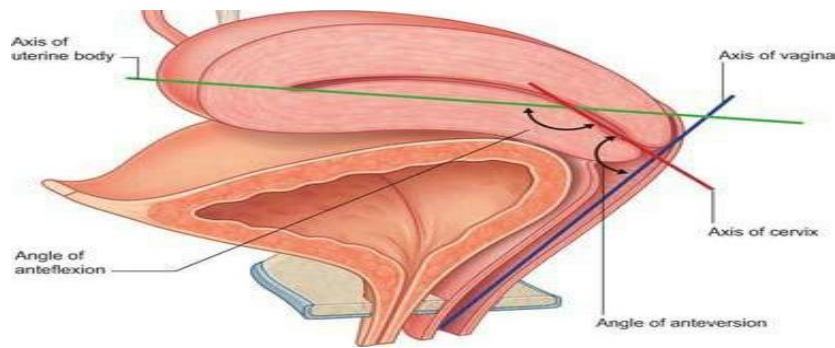
Situation : in the lesser pelvis between urinary bladder and rectum



Normal position of uterus

Angle of anteversion : it is the forward angulation between long axis of cervix and long axis of vagina. It is about 90°.

Angle of anteflexion : It is the forward angulation between long axis of the body of uterus and long axis of the cervix. It is about 120°.



Parts of the uterus

It is divided into

1. Body - upper expanded part
2. Cervix – lower cylindrical part
3. Isthmus – constriction between body and cervix

Body : Include

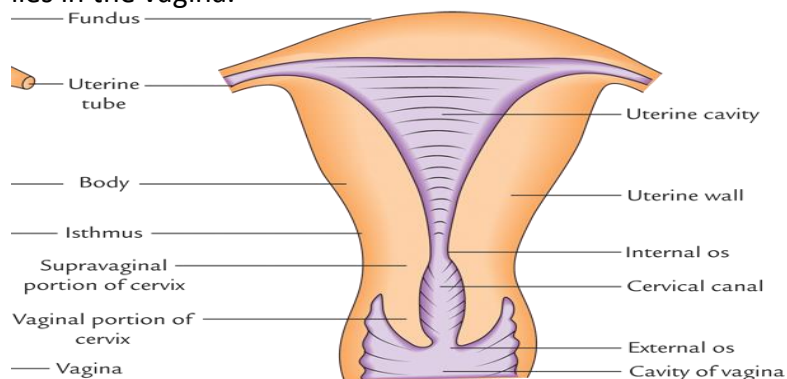
1. Funds : It is the convex part present above the opening of uterine tube.
2. Two surfaces : Anterior and posterior
3. Two lateral borders

Anterior surface : It is flat and it reflected on the upper surface of the bladder to form uterovesical pouch.

Posterior surface : Convex and it covered by peritoneum. This peritoneum forms the anterior wall of rectouterine pouch.

Two lateral borders : It is rounded and give attachments to broad ligament , uterine tube and round ligament.

Cervix : It is the lower cylindrical part. It is about 2.5cm long . It project in to the anterior wall of the vagina, dividing in to two parts .Supravaginal part : It lies above the vagina.Vaginal part : It lies in the vagina.



Cavity of uterus

Triangular in coronal section. It communicate with cervical canal through internal os. Vaginal part of cervix contain a depression called external os, through which cervical canal communicate with vagina.

Ligaments of uterus

1. Broad ligament
2. Round ligament of uterus
3. Transverse cervical ligament
4. Uterosacral ligament

5. Anterior ligament is called uterovesical fold of peritoneum
6. Posterior ligament is called rectovaginal fold of peritoneum

Supports of the uterus

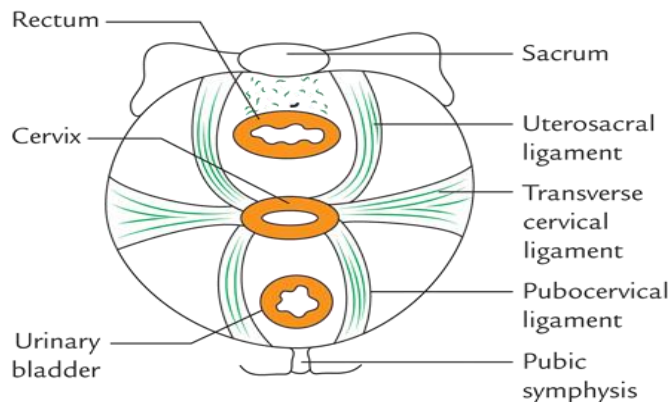
1. Primary supports : Muscular, fibromuscular and visceral
2. Secondary supports : Broad ligament, anterior and posterior ligaments
3. Visceral : Urinary bladder, vagina and uterine axis

I. Muscular supports

1. **Pelvic diaphragm** : It is a muscular structure formed by levator ani and coccygeus muscles. It forms the floor of pelvis and supports all the pelvic organs.
2. **Perineal body** : it is a wedge shaped mass of fibromuscular tissue. Situated 1.25 cm in front of anal margin. It gives attachments to two superficial and deep transverse perineal muscles.
3. **Urogenital diaphragm** : it is a triangular muscle formed by sphincter urethrae and deep transverse perineal muscles. It supports the uterus as well as the vagina.

II. Fibromuscular

1. **Pubo cervical ligaments** : It connects cervix to the posterior surface of pubis
2. **Transverse cervical ligaments** : It is fan shaped fibro muscular band it connects lateral aspect of cervix to the lateral pelvic wall.
3. **Utero sacral ligaments** : extends from cervix to sacrum.



4. **Round ligament of the uterus** : it is about 10 to 20cm. it lies between two layers of broad ligaments. It begins at the lateral angle of uterus and runs forward and laterally enters deep inguinal ring and transverse the inguinal canal and comes out through superficial inguinal ring and divides into numerous fibrous bands and joins with fibroareolar tissue of labia majora.

Arterial supply : uterine artery and ovarian artery

Venous drainage : uterine vein drains into the internal iliac vein

Nerve supply : sympathetic T12 and L1

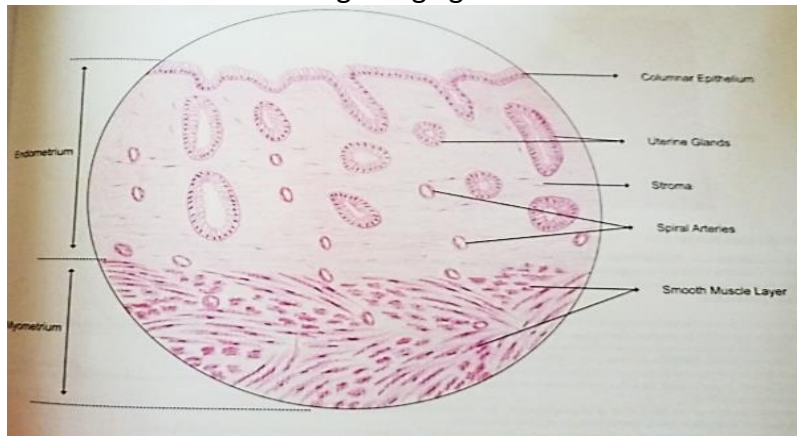
Parasympathetic S2 to S4

Microscopic structure

It has 3 layers - outer perimetrium, middle myometrium and inner endometrium.

Endometrium : lined by columnar epithelium. Stroma contains simple tubular glands and spiral arteries. Endometrium is divided into : outer parts (stratum) functionalis which is shed during the menstrual phase. Inner parts (stratum) basalis which help in regeneration of the shed

endometrium. **Proliferative phase** – under the influence of estrogen endometrium increases in thickness, gland elongate, spiral artery reaches upto the surface and stromal cells increases in number. **Secretory phase** – under the influence of progesterone, the endometrium further increases in thickness. Glands are dilated and tortuous giving a saw-toothed appearance in section. **Myometrium** – three ill-defined layers of smooth muscles. Blood vessels run in between the muscle fibers forming living ligatures. **Perimetrium** – made of connective tissue coverings.



Applied anatomy

1. Prolapse of uterus : It supports of uterus become weak due to any cause uterus descends in vagina is called prolapse of uterus. It may be loss of tone muscles pelvic diaphragm, perineal tears and repeated child birth
2. Surgical removal of uterus called hysterectomy
3. Cervical carcinoma

VAGINA

It is a female organ of copulation and forms the lower part of the birth canal.

Size : length anterior wall is about 7.5cm and posterior wall is about 9cm.

Width – upper end 5cm and lower end 2.5cm.

Shape in cross section : 'H' shape in lower 1/3rd, transverse slit in middle 1/3rd and round in upper 1/3rd.

Fornices of vagina

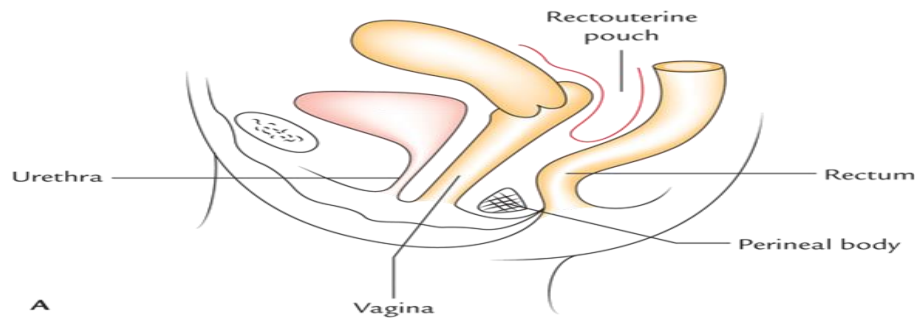
The circular area of vaginal lumen around the cervix is called fornix of vagina. It is divided into four parts . anterior fornix, posterior fornix and two lateral fornix.

Relations

Anteriorly : upper half by urinary bladder and lower half by urethra.

Posteriorly : upper ¼ by rectouterine pouch, middle 2/4 by ampulla of rectum and lower 1/4 by Perineal body.

Laterally : ureters, levator ani and pelvic fascia, urogenital diaphragm,



Arterial supply : vaginal , internal iliac , internal pudental, middle rectal and inferior vesical arteries.

Venous drainage : pampiniform plexus of vein drains into vaginal vein drains into internal iliac Veins.

Nerve supply : upper 2/3rd sympathetic – L1 and L2
Parasympathetic – S2 and S3
Lower 1/3rd by pudental nerve

Applied anatomy

Vaginitis : infection of vagina

BREAST OR MAMMARY GLAND

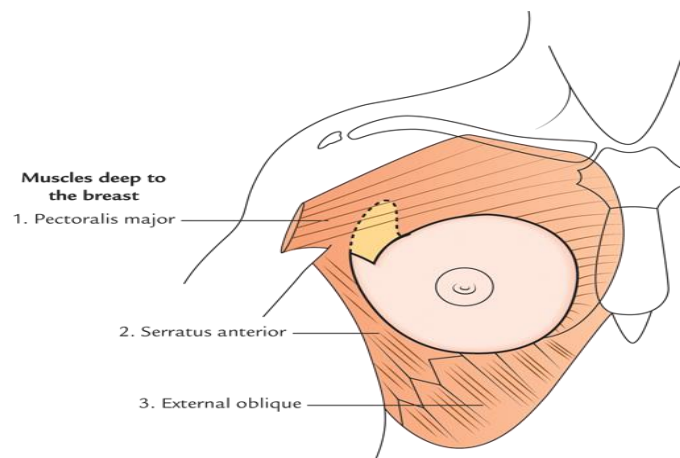
It is a modified sweat gland. It is an accessory organ of the female reproductive system.

It produces nutrition to the newborn in the form of milk.

Situation : it lies in the superficial fascia of the pectoral region. A small extension is called axillary tail of spence and it extend into the deep fascia.

Extend : second to sixth ribs

Deep relations : deep fascia or pectoral fascia, pectoralis major, serratus anterior and external oblique abdominis muscles

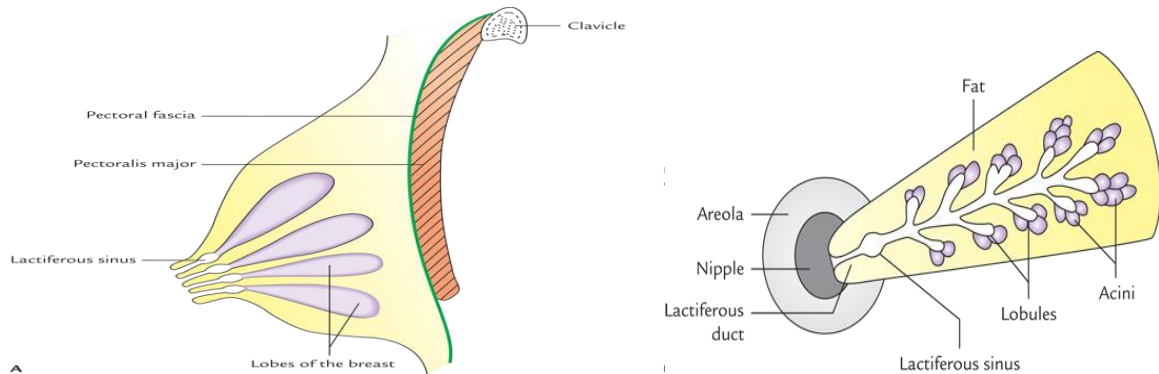


Structure of breast

It is divided into skin, paranchyma and stroma.

1. **Skin** : it covers gland and consists of

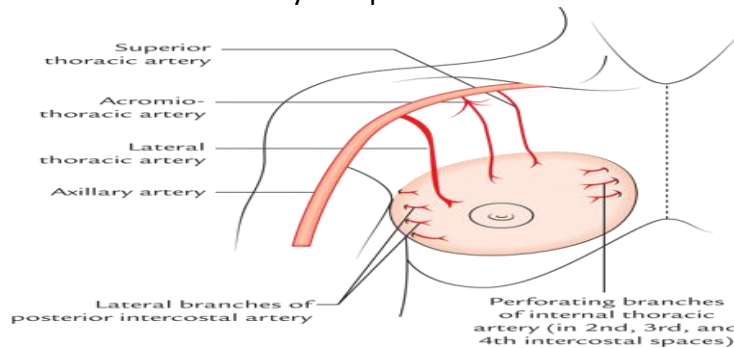
- a. **Nipple** : it is a conical projection, present below the centre of the breast. It is pierced by 15 to 20 lactiferous ducts.
- b. **Areola** : the skin surrounding the base of the nipple is pigmented and forms a circular area called the areola.
2. **Paranchyma** : it is a compound tubuloalveolar gland which secrete milk. The gland consists of 15 to 20 lobes. Each lobe is a cluster of alveoli. Alveoli drained by lactiferous duct. Lactiferous duct converge towards the nipple and open it.
3. **Stroma** : it forms the supporting frame work of gland. It is composed of partly fibrou and partly fatty. The fibrous septa forms the suspensory ligament of cooper, which anchor the skin and gland to the pectoral fascia.



Arterial supply : it is supplied by internal thoracic, lateral thoracic, superior thoracic and posterior intercostal arteries.

Venous drainage : the veins follow the arteries.

1. Superficial veins drains into the internal thoracic vein and into the superficial vein of the lower part of neck.
2. Deep vein drains into the axillary and posterior intercostal veins.



Nerve supply

It is supplied by 4th to 6th intercostal nerve.

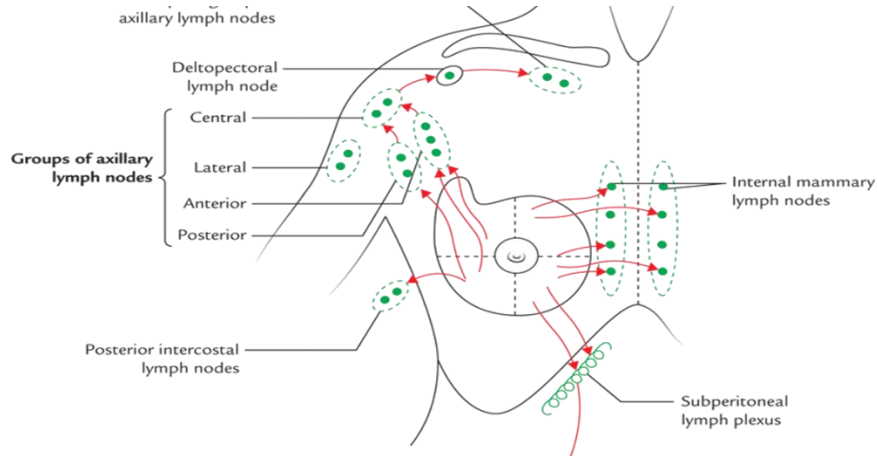
Lymphatic drainage

The lymph nodes draining the breast are axillary lymph nodes, internal mammary lymph node, supraclavicular nodes, deltopectoral lymph nodes, posterior intercostal lymph nodes, subdiaphragmatic lymph nodes and subperitoneal lymph nodes.

Lymphatic vessels

- Superficial lymphatic : it drains the skin over the breast except for the nipple and areola. These lymphatic drains into axillary, internal thoracic, supraclavicular and deltopectoral lymph nodes.
- Lymph from areola, nipple and paranchyma drains into axillary lymph nodes

- About 75% of the lymph from the breast drains into the axillary lymph nodes.
- About 20% into the internal thoracic lymph nodes.
- About 5% into the posterior intercostal lymph nodes.

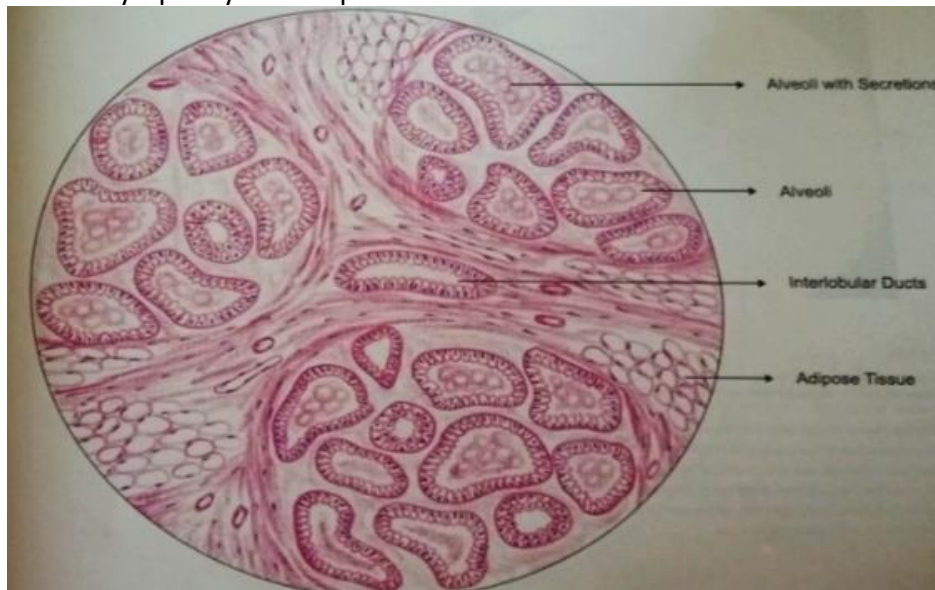


Microscopic structure

Inactive mammary gland : contain more connective tissue and less glandular tissue

Parenchyma ; consist of less glandular tissue, poorly developed alveoli . More connective tissue and adipose tissue present.

Active mammary gland : Parenchyma : glandular tissue more prominent , alveoli is highly developed and lined by cuboidal epithelium. Interlobular duct lined by cuboidal epithelium, Stroma : less connective tissue and adipose tissue, Intralobular connective tissue is less and contain lymphocytes and plasma cells.



Applied anatomy

- Cancer cell may infiltrate the suspensory ligaments. The breast then become fixed. Contraction of the ligaments can causes retraction of skin.
- Infiltration of lactiferous ducts and their consequent fibrosis can cause retraction of the nipple'

- Obstruction of superficial lymphatic vessels by cancer cells may causes orange colour appearance in the skin.

Difference between

Testis	Ovary
Male gonad	Female gonad
Oval shape	Almond shape
Produce testosterone and spermatozoa	Produce oocytes
Suspended in scrotum by spermatic cord	Lies in the ovarian fossa

Fallopian tube	Vas deferense
It is the part of female reproductive system	It is the part of male reproductive system
It present in the abdominal cavity	It from the scrotal sac to abdominal cavity
It is about 10cm long	It is about 45cm long
It further connect with uterus	It further connect with urethra
It carry ovum from the ovary to uterus	It carry sperm from epididymis to ejaculatory duct

Questions

1. Parts of female reproductive system. Explain uterus in details.
2. Supports of uterus
3. Fallopian tube
4. Breast or mammary gland
5. Ovary
6. Vagina
7. Parts of male reproductive system . Explain testis in details
8. Prostate
9. Epididymis
10. Vasdeferense

Difference between

11. Ovary and testis
12. Vasdeferense and fallopian tube
13. Male and female reproductive system
14. List the parts of male reproductive system
15. List the parts of female reproductive systemm